

REG.NO:

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GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B.E. Degree Examinations – Nov. / Dec. 2025

ELECTRONICS AND COMMUNICATION ENGINEERING

Fifth Semester

22LPC502 – Digital Signal Processing

Regulations 2022

Time : 3 Hours

Maximum Marks : 100

Answer ALL Questions

Part - A (10 x 2 = 20 Marks)

- | | M | CO | BT |
|---|---|----|----|
| 1. What is meant by twiddle factor and state its properties? | 2 | 1 | U |
| 2. What is meant by bit reversal and in place computation as applied to FFT? | 2 | 1 | U |
| 3. By Impulse Invariance method, obtain the digital transfer function and differential equation of the analog filter $H(s) = \frac{1}{s+1}$. | 2 | 2 | A |
| 4. What is meant by prewarping? Give remedial measure to prevent this. | 2 | 2 | AN |
| 5. What is the necessary and sufficient condition for the impulse response $h(n)$ of an FIR filter to be linear phase? | 2 | 3 | R |
| 6. What are the desirable characteristics of the frequency response of window function? | 2 | 3 | R |
| 7. Distinguish between truncation and rounding of binary digits using relevant examples. | 2 | 4 | U |
| 8. Define signal scaling? | 2 | 4 | R |
| 9. Define multi rate signal processing? | 2 | 5 | R |
| 10. Consider the given discrete signal $x(n) = \{1, -1, 0.5, 1\}$. Find its interpolated version with a factor $I = 2$. | 2 | 5 | A |

Part - B (5 x 16 = 80 Marks)

- | | | | |
|--|----|---|---|
| 11. a) i) Find the 8-point DFT of the given sequence using radix-2 DIT-FFT Algorithm.
$x(n) = \{1, 0, 1, 0, 1, 0, 1, 0\}$ | 10 | 1 | A |
| ii) State the convolution property of N-point DFT and also perform the circular convolution between the following two discrete sequences $x(n) = \{1, 2, 0, -1\}$ and $p(n) = \{1, 0, -1, 2\}$. | 06 | 1 | A |
| (OR) | | | |
| b) i) Compute the response of a discrete system to the input $x(n) = \{1, 2, -1, -2, 3, -2\}$ and impulse response $h(n) = \{1, 2\}$ using Overlap - Add method. | 10 | 1 | A |

- ii) Compute N-point DFT of the following sequence using radix-2 DIF-FFT method. 06 1 A

$$x(n) = \sin\left(\frac{n\pi}{2}\right), 0 \leq n \leq 3$$

$$= 0, \text{ otherwise}$$

12. a) Design a Butterworth Low Pass Filter satisfying the following constraints: 16 2 A

$$0.707 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq 0.5\pi$$

$$|H(e^{j\omega})| \leq 0.2, \quad 0.75\pi \leq \omega \leq \pi$$

Use Bilinear Transformation Technique and assume that $T=1$ sec.

(OR)

- b) Design an Analog Chebyshev filter for the following specifications: 16 2 A
- pass band attenuation = 0.89, stop band attenuation = 0.2
 - pass band edge frequency = 30 Hz,
 - stop band edge frequency = 75 Hz.

13. a) Design an FIR filter for the ideal frequency response using Hamming Window with $N=7$. 16 3 A

$$H_d(e^{j\omega}) = e^{-j3\omega}, \quad -\frac{\pi}{8} \leq \omega \leq \frac{\pi}{8}$$

$$= 0, \quad \frac{\pi}{8} \leq |\omega| \leq \pi$$

(OR)

- b) Determine the filter coefficient $h(n)$ of length $M=15$ obtained using frequency sampling method, given that the frequency response is specified as follows: 16 3 A

$$H\left(\frac{2\pi k}{15}\right) = 1, \quad k = 0, 1, 2, 3, 4$$

$$= 0.4, \quad k = 5$$

$$= 0, \quad k = 6, 7$$

14. a) i) Explain the characteristics of limit cycle operations with respect to the system described by the difference equation $y(n) = 0.95 y(n-1) + x(n)$. Assume that No of bits used for Quantization of the product through Rounding is 4 bits and $x(n) = 0.75$ for $n = 0$; $x(n) = 0$ for $n \neq 0$; Determine the dead band of the filter. 10 4 A

- ii) Bring out the differences between fixed point and floating-point arithmetic.

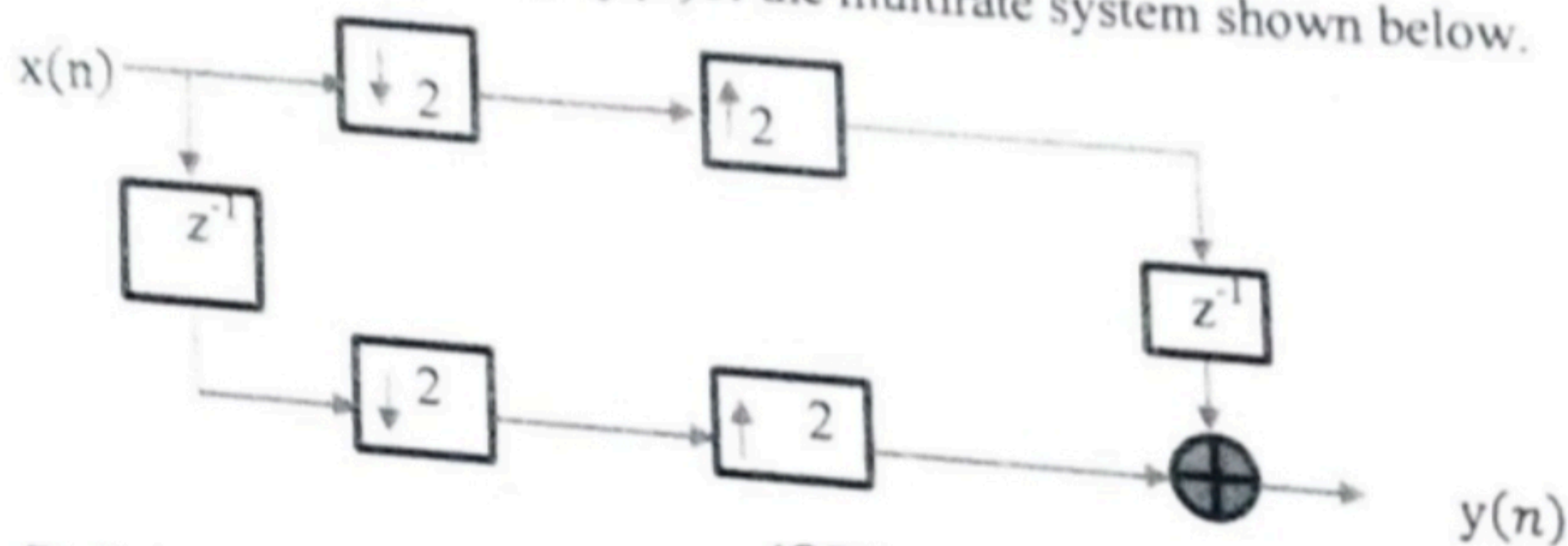
(OR)

- b) i) The output signal of an ADC is passed through a first order lowpass filter with transfer function given by $H(z) = \frac{(1-a)z}{z-a}$ for $0 < a < 1$. Calculate the steady state output noise power due to quantization at the output of the digital filter. 10 4 A

- ii) Describe the various formats of the fixed-point representation of binary numbers. 06 4 U

15. a) i) Provide a detailed explanation of the concept of decimation by a factor 'D' in signal processing. Derive the corresponding output spectrum of the decimated signal. 10 5 U

- ii) Find the output expression $y(n)$ of the multirate system shown below. 06 5 A



(OR)

- b) i) Define an adaptive filter. With the aid of a clear circuit diagram, explain its applications (at least 3) in signal processing. 10 5 U

- ii) Explain the concept of sampling rate conversion by the factor (I/D) and obtain its output spectrum. 06 5 /

BT	R	U	A	AN	E	C
%	24	08	36	32	-	-